

Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | Context: North India Competitive Exam Aspirants (Hindi-Medium)

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MMLU's output ontology is a four-class A/B/C/D classification [Q83, Q44] with classification accuracy as the sole performance criterion [Q34, Q23]. Dataset analysis confirms a single ClassLabel field (0–3) with no rationale field. The deployment requires a fundamentally richer output taxonomy: a verdict (sahi/galat) plus a substantive Hindi-language explanation in standard Manak Hindi [elicitation Q3]. MMLU's design provides zero signal on the model's capacity to produce explanatory rationale, let alone in Hindi. The benchmark's stated rationale for the simple format ('a simple-to-evaluate test that measures classification accuracy' [Q23]) is internally coherent but structurally incompatible with the deployment's pedagogical-feedback requirement.

ID	Evidence
Q3	"We find that while most recent models have near random-chance accuracy, the very largest GPT-3 model improves over random chance by almost 20 percentage points on average." (p.1)
Q23	"Consequently, we instead create a simple-to-evaluate test that measures classification accuracy on multiple choice questions." (p.3)
Q34	"To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)
Q44	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)
Q83	"Models are fine-tuned to predict one of four classes using the UnifiedQA MCQ questions and using our dev+val set. We test on our multitask test set." (p.12)

Strengths

- Verdict component (correct vs. incorrect) is a partial match — the A/B/C/D classification can be reduced to a binary correct/incorrect signal that does align with one component of the deployment output (the sahi/galat label) [Q83, DATASET concern 3].

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Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values

OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output labels (A/B/C/D) [Q83] partially match the deployment's required correctness verdict but entirely omit the explanation component. Deployment requires Hindi explanatory feedback per elicitation Q3.

OO-2: Missing categories: free-form Hindi explanation, distractor analysis (why each wrong option is wrong), and reference to canonical Indian sources (NCERT, Laxmikant). None are represented in MMLU's output schema.

OO-3: Output categories themselves are content-neutral (letter labels), but the underlying scoring rule treats US-anchored answer keys (e.g., moral_scenarios under US 2020 norms [DATASET-D18, D19]) as correctness ground truth, encoding non-regional values.

OO-4: A stakeholder-driven output taxonomy redesign is required: minimum {verdict_label, hindi_explanation_text, optional_distractor_analysis} with a separate rubric for explanation quality. No existing benchmark covers this — the rubric gap is documented as unresolved [WEB-21].

OO-5: Structural and content validity are severely harmed: MMLU's output structure measures only one of two required output components and applies decision rules anchored in US norms.

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Q62	"Importantly, the format of our evaluation is not identical to the format in which information is acquired during pretraining." (p.8)
Q83	"Models are fine-tuned to predict one of four classes using the UnifiedQA MCQ questions and using our dev+val set. We test on our multitask test set." (p.12)
D18	"according to ordinary moral standards in the US as of 2020? Scenario 1 — The bomb shelter was filling up, but I made sure my grand piano got a spot."
D19	"according to ordinary moral standards in the US as of 2020? Scenario 1 — When I went through the road block I gave the policeman my brothers ID."
WEB-13	MILU derives labels from official Indian competitive exams — implicit Indian-stakeholder validity that MMLU lacks (https://aclanthology.org/2025.naacl-long.507/)
WEB-15	ParamBench uses two-tier annotation team with Indian-context expert review — contrast model for what MMLU lacks (https://arxiv.org/abs/2508.16185)
WEB-21	No Hindi exam-prep explanation-quality rubric exists (https://arxiv.org/abs/2508.19831)

Information Gaps

- No published rubric for Hindi exam-prep explanation quality [WEB-21] — cannot quantify the magnitude of the output-ontology gap.

Requires Expert Verification

- Indian exam-prep pedagogues need to define an explanation-quality rubric (factual accuracy, distractor analysis, Manak Hindi register, NCERT alignment) for the deployment.

Remediation

Gap 1: Label-only A/B/C/D output cannot evaluate the required Hindi explanatory feedback.

Recommendation: Commission a Hindi exam-prep explanation-quality rubric (factual accuracy, distractor analysis, Manak Hindi register, NCERT alignment) with Indian SME panel — this is a documented gap in current literature [WEB-21] and must be built bespoke for the deployment.

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